

Email DNS Records: Configuring MX, SPF, DKIM & DMARC

Introduction

In the previous tutorial, *A Records and Subdomains: Connecting Names to Servers*, we used DNS to point a name to the right server for a website.

Email also relies on DNS, but it adds an extra requirement: other mail servers must confirm that messages from your domain are legitimate. Without this, mail may work, but often ends up in spam or is rejected.

Most email issues fall into two situations:

- **Incoming email does not arrive**, because senders cannot find the correct mail server for your domain.
- **Outgoing mail goes to spam** because receivers cannot confirm that your domain is allowed to send those messages.

In this tutorial, we will check and configure the DNS records that solve both:

- **MX** for receiving email
- **SPF** and **DKIM** for proving that your outgoing email is legitimate
- **DMARC** for setting a clear policy and receiving reports

By the end, you will be able to check, add, and correct MX, SPF, DKIM, and DMARC records for your domain.

This tutorial applies whether you use ServeTheWorld mail hosting, Google Workspace, Microsoft 365, or a mix of services.

Key Terminology

- **Mail server** — A system that sends or receives email
- **MX record** — A DNS record that points incoming email to the right mail server
- **TXT record** — A DNS record that stores text, often used for email authentication
- **Selector** — The short label at the start of a DKIM record name, such as `default` or `selector1`
- **Priority** — The number that determines which mail server receives messages first (lower numbers = higher priority)

Deliverability: Why You Need Email DNS Records

Email servers do not automatically trust each other. Every message is checked before it is accepted.

Before delivering an email, a receiving server asks:

- Is this server allowed to send mail for this domain?
- Has the message been altered in transit?
- What policy applies if authentication fails?

If these questions are not answered clearly, your email may bounce back to the sender with a failed delivery notice, be filtered as spam, or rejected.

The MX Record

When someone emails `contact@yourdomain.com`, the sending mail server looks up your MX records to find out where to deliver the message.

An MX (Mail Exchange) record specifies which server receives incoming email. Without one, your domain may not receive email reliably.

A basic MX record looks like:

```
yourdomain.com  MX  10  mail.yourprovider.com
```

Priority Levels

When multiple MX records exist, priority determines order. **Lower numbers indicate higher priority.**

Example:

```
yourdomain.com  MX  10  mail1.yourprovider.com
yourdomain.com  MX  20  mail2.yourprovider.com
```

Mail is delivered to priority 10 first. If that server is unavailable, priority 20 is used. This method creates an automatic failover.

Check Your MX Records

```
nslookup -type=MX yourdomain.com
```

You will usually see one or more "mail exchanger" lines. Each entry has a preference value.

Example:

```
Non-authoritative answer:
yourdomain.com MX preference = 10, mail exchanger = mx0101.titan.email
yourdomain.com MX preference = 20, mail exchanger = mx0102.titan.email
```

Here, mail is sent to the priority 10 server first, with priority 20 acting as backup.

If you see no MX records, or they point to the wrong provider or an old mail server, incoming mail can fail.

Add MX Records

Primary MX:

- Record type: MX
- Name/Host: @
- TTL: 3600
- Priority / Preference: 10
- Mail server / Exchange: mx0101.titan.email

Secondary MX:

- Record type: MX
- Name/Host: @
- TTL: 3600
- Priority / Preference: 20
- Mail server / Exchange: mx0102.titan.email

SPF (Sender Policy Framework)

SPF is a public list of which servers are allowed to send email on behalf of your domain. When you send a message, many receiving mail servers check SPF to confirm it came from an approved system rather than an impersonator.

SPF is stored as a TXT record.

Example:

```
v=spf1 include:_spf.titan.email -all
```

This record means:

- Use SPF version 1
- Authorise servers used by your provider
- Reject all other sources (`-all`)

Only one SPF record is allowed per domain. If you publish multiple SPF policies, some mail servers treat SPF as invalid.

SPF Enforcement Levels

At the end of the record, you see a qualifier:

1. **-all**: Reject email from servers that are not authorised
2. **~all**: Mark email as suspicious
3. **?all**: Neutral

For most domains, use `-all`. If testing a new configuration, temporarily use `~all`, then change to `-all` once confirmed.

Check Your SPF Record

```
nslookup -type=TXT yourdomain.com
```

A domain can have several TXT records. Look for the one that begins with: `v=spf1`

That entire line is your SPF policy.

Common SPF Examples

ServeTheWorld:

```
v=spf1 include:_spf.titan.email -all
```

Google Workspace:

```
v=spf1 include:_spf.google.com -all
```

Microsoft 365:

```
v=spf1 include:spf.protection.outlook.com -all
```

Multiple services (ServeTheWorld + SendGrid mail service):

```
v=spf1 include:_spf.titan.email include:sendgrid.net -all
```

Add an SPF Record

- Record type: `TXT`
- Name/Host: `@`
- TTL: `3600`
- Value / Content: `v=spf1 include:_spf.titan.email -all`

If you use additional senders, add them to the same SPF record rather than creating a second SPF policy.

DKIM: Verifying Message Integrity

DKIM (DomainKeys Identified Mail) helps prove that an email really came from your domain and was not modified along the way. It does this by adding a cryptographic signature to each outgoing message and publishing a matching public key in DNS that receiving servers can use to verify the signature.

Your mail server signs the message using a private key. The receiving server retrieves the corresponding public key from DNS and verifies the signature.

If the signature cannot be verified, the message may be flagged or rejected by the receiving system.

What is a DKIM Selector?

A selector is a short label in the DKIM record name, used to identify which key is in use (like `default` or `selector1`). Your email provider will tell you which selector to use and what value to publish.

Check Your DKIM Record

```
nslookup -type=TXT selector._domainkey.yourdomain.com
```

If you do not know the selector, check your provider's control panel or setup instructions. You can try `default` as a quick check, but do not assume it is correct:

```
nslookup -type=TXT default._domainkey.yourdomain.com
```

A DKIM record usually contains a value that starts with `v=DKIM1` and includes a long `p=` key.

Add a DKIM Record

- Record type: **TXT**
- Name/Host: **selector._domainkey** (most DNS panels add **.yourdomain.com** automatically)
- TTL: **3600**
- Value / Content: the DKIM value provided by your mail provider, usually starting with **v=DKIM1** and including a long **p=** key

DMARC: Policy and Reporting

DMARC (Domain-based Message Authentication, Reporting and Conformance) builds on SPF and DKIM by defining policy: **what should receiving servers do if authentication fails?**

It also enables reporting to monitor authentication results and identify configuration errors or spoofing attempts. Interpreting DMARC reports is outside the scope of this tutorial. Here, we focus only on publishing a correct policy record.

Check Your DMARC Record

```
nslookup -type=TXT _dmarc.yourdomain.com
```

A DMARC record starts with **v=DMARC1**. A common, safe starting point looks like this:

```
v=DMARC1; p=none; rua=mailto:admin@yourdomain.com
```

DMARC Policy Progression

A practical rollout is gradual:

1. Start with **p=none** and monitor for 1–2 weeks.
2. Then, move to **p=quarantine** for another 1–2 weeks.
3. Finally, move to **p=reject** once all legitimate mail authenticates correctly.

Add a DMARC Record

- Record type: **TXT**
- Name/Host: **_dmarc** (most DNS panels add **.yourdomain.com** automatically)
- TTL: **3600**
- Value / Content: **v=DMARC1; p=none; rua=mailto:admin@yourdomain.com**

After saving, you should see a TXT record named **_dmarc** (or **_dmarc.yourdomain.com**) with a value that begins with **v=DMARC1**.

How to Add or Edit Records in ServeTheWorld DNS Manager

When you see blocks like "Record type / Name/Host / TTL / Value/Content" below, treat them as "fill in these fields in your DNS editor". Complete sample record recipes are provided in the next section.

1. Open <https://my.servetheworld.net> and log in.
2. In the control panel, open **Domains** → **My Domains**.
3. Select your domain, then choose **Manage** → **Manage DNS**.
4. From there, add a new record.
5. Choose record type (MX or TXT).
6. Fill in Name/Host, TTL, and Value/Content.
7. Save, then re-check with the commands above.

DNS changes usually take effect within one hour. Some networks cache results longer. Allow up to 24–48 hours for full propagation.

Quick Reference: Record Recipes

Complete examples for common email providers:

ServeTheWorld

MX Records

Common settings:

```
Record type: MX
Name/Host: @
TTL: 3600
```

Servers:

```
Priority: 10
Value: mx0101.titan.email

Priority: 20
Value: mx0102.titan.email
```

SPF

Record type: TXT
Name/Host: @
TTL: 3600
Value: v=spf1 include:_spf.titan.email -all

DKIM

Record type: TXT
Name/Host: default._domainkey
TTL: 3600
Value: (Provided by ServeTheWorld, begins with v=DKIM1; and includes a long p= key)

DMARC

Record type: TXT
Name/Host: _dmarc
TTL: 3600
Value: v=DMARC1; p=none; rua=mailto:admin@yourdomain.com

Google Workspace

MX Records

Common settings:

Record type: MX
Name/Host: @
TTL: 3600

All Google mail servers:

Priority: 1
Value: aspmx.l.google.com

Priority: 5
Value: alt1.aspmx.l.google.com

Priority: 5

Value: `alt2.aspmx.l.google.com`

Priority: `10`

Value: `alt3.aspmx.l.google.com`

Priority: `10`

Value: `alt4.aspmx.l.google.com`

SPF

Record type: `TXT`

Name/Host: `@`

TTL: `3600`

Value: `v=spf1 include:_spf.google.com -all`

DKIM

Record type: `TXT`

Name/Host: `google._domainkey`

TTL: `3600`

Value: (Generated in Google Workspace Admin under DKIM settings)

DMARC

Record type: `TXT`

Name/Host: `_dmarc`

TTL: `3600`

Value: `v=DMARC1; p=none; rua=mailto:admin@yourdomain.com`

Microsoft 365

MX Record

Record type: `MX`

Name/Host: `@`

TTL: `3600`

Priority: `0`

Value: `yourdomain-com.mail.protection.outlook.com`

SPF

```
Record type: TXT
Name/Host: @
TTL: 3600
Value: v=spf1 include:spf.protection.outlook.com -all
```

DKIM

Microsoft 365 uses two selectors.

```
Record type: TXT
Name/Host: selector1._domainkey
TTL: 3600
Value: (Generated in Microsoft 365 admin centre)
```

```
Record type: TXT
Name/Host: selector2._domainkey
TTL: 3600
Value: (Generated in Microsoft 365 admin centre)
```

DMARC

```
Record type: TXT
Name/Host: _dmarc
TTL: 3600
Value: v=DMARC1; p=none; rua=mailto:admin@yourdomain.com
```

Troubleshooting Checklist

If incoming mail does not arrive, check that:

- MX records exist and point to the correct mail server (no spelling mistakes)
- Priorities are correct if using multiple MX records (lower = higher priority)

If outgoing mail often lands in spam, check that:

- One SPF record exists and includes all sending services
- Your SPF record does not exceed the 10-lookup limit (SPF allows only 10 DNS lookups; too many `include` entries can cause mail to fail).
- DKIM is enabled, and the TXT record exists under the correct selector
- DMARC is set to `p=none` until SPF and DKIM work correctly

If results differ across networks:

- Wait a little and run the lookups again (DNS caching can cause temporary inconsistencies)
- Make sure you edited the correct DNS zone
- Confirm you did not accidentally publish multiple SPF records

Conclusion

Once configured, these records work quietly in the background: MX ensures mail reaches you, SPF confirms authorised senders, DKIM protects message integrity, and DMARC enforces policy.

Together, they protect your domain's reputation and help ensure your messages reach the inbox.

In the next tutorial, we'll explore SRV records used by VoIP systems, Microsoft Teams, and game servers.